Requirements Table

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #07 — Campus & Academic Operations

Hostel Maintenance & Issue Ticketing System

Hostel administration requires a geo-tagged ticketing pipeline to report plumbing, electrical, and carpentry issues, dispatch maintenance staff, track SLA timelines, and record resident closure sign-offs.

Target Stakeholders / Actors: Hostel Resident, Maintenance Warden (Maintenance Staff identified as an additional actor — see Use-Case Diagram).

Requirement Interpretation

FR-001 and NFR-001 are taken directly from the assigned problem statement and are preserved unchanged in intent and wording. FR-002 through FR-005 were derived to cover the remaining stages of the ticketing pipeline named in the problem context: dispatch/assignment (FR-002), SLA-timestamped status tracking (FR-003), completion evidence (FR-004), and resident closure sign-off (FR-005). NFR-002 was added as a genuinely non-functional, measurable constraint on the pipeline's geo-tagging accuracy, distinct from the escalation-alert constraint already covered by NFR-001.

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to the warden queue. Fail: Incomplete form accepted without a room tag.	Establishes the core intake mechanism for all reported plumbing, electrical, and carpentry issues.
FR-002	Functional	The system shall enable the Maintenance Warden to assign an open ticket to a maintenance staff member based on issue category and staff availability.	High	Pass: Selected ticket status changes from "Open" to "Assigned" and the named staff member appears on the ticket record. Fail: Ticket remains unassigned or is assigned without an identified staff member.	Enables dispatch of the correct maintenance staff, a core stage of the ticketing pipeline.
FR-003	Functional	The system shall allow assigned maintenance staff to update a ticket's status (Assigned, In Progress, Work Completed) and shall timestamp every status change against the ticket's SLA clock.	High	Pass: Each status change is saved with a timestamp and is visible on the ticket's SLA timeline. Fail: A status change is accepted without a corresponding timestamp record.	Provides the SLA-timeline tracking needed to measure and later escalate against the 24-hour target.
FR-004	Functional	The system shall require maintenance staff to attach photo evidence of completed work before a ticket can be marked "Work Completed".	Medium	Pass: Ticket transitions to "Work Completed" only after at least one completion photo is attached. Fail: Ticket transitions to "Work Completed" with zero attached photos.	Gives the resident verifiable evidence to review before sign-off and discourages unverified closures.
FR-005	Functional	The system shall enable the Hostel Resident to review completed work and record a closure sign-off (Confirm Closure or Reject and Reopen) before the ticket is marked "Closed".	High	Pass: Ticket status becomes "Closed" only after the resident records a Confirm Closure decision. Fail: Ticket is marked "Closed" without a resident sign-off record.	Ensures the resident, as the affected stakeholder, formally confirms the issue is resolved.
NFR-001	Non-Functional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Guarantees high-priority issues are not silently missed once the SLA window lapses.
NFR-002	Non-Functional	The system shall capture a geo-location tag (hostel block and room coordinates) for every ticket, accurate to within 10 metres.	Medium	Pass: Audit of 100 sampled tickets shows ≥99% recorded within 10 metres of actual room location. Fail: >1% of sampled tickets exceed the 10-metre band.	Supports the geo-tagged pipeline required by the problem statement and speeds staff dispatch.

Summary: Total Functional Requirements: 5 (FR-001–FR-005) | Total Non-Functional Requirements: 2 (NFR-001–NFR-002)